

Insulin

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

This insulin assay is performed using a 2-site electrochemiluminescent immunoassay on the Roche automated platform.¹ This test could be effectively used for those patients not receiving exogenous insulin or those receiving unmodified human insulin. Test 004333, Insulin is highly specific for human insulin, as shown in the table below. This test does not react, for instance, with several of the insulin analogs used to treat diabetes as indicated in Table 1 below. The efficacy of each test depends on the patient treatment and goal of the ordering physician. It is important that the physician ordering and interpreting the results understands the specificity of the test.

Table 1: Cross reactivity to insulin analogs¹⁻³

- Human insulin: 100%
- Insulin aspart: <1%
- Insulin glargine: <1%
- Insulin lispro: <1%
- Insulin glulisine: <1%
- Insulin detemir: <1%

Notes:

- Circulating antibodies to insulin, both autoantibodies and antibodies to therapeutic insulin, can interfere with the immunoassays for insulin.^{4,5}

- As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.¹ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.¹ The test contains additives that minimize these effects.

Custom Additional Information

Insulin is a peptide hormone with a molecular weight of approximately 6000 daltons. Serum insulin determinations may be performed on patients with symptoms of hypoglycemia^{4,5} where insulinoma is a possible etiology. Patterns of secretion by insulinomas may be sporadic or constant. Insulinoma is a rare functional tumor of pancreatic origin that secretes insulin inappropriately, causing hypoglycemia. Whipple originally described the tumor and classic diagnostic criteria called "Whipple's Triad". The criteria included:

- 1) neurologic symptoms of hypoglycemia
- 2) low glucose (<50 mg/dL) which are corrected immediately by giving glucose or carbohydrate.

Today, as described above, we would add inappropriately elevated insulin levels during an episode of hypoglycemia. Usually hypoglycemia is induced with a fast or exercise, and the diagnostic criteria may include a 48 to 72 hour fast with insulin levels greater than 5 μ U/mL during a hypoglycemic episode.⁴

Hypoglycemia can, however, also be facilitated intentionally or unintentionally (factitious hypoglycemia),^{4,5} C-peptide levels along with insulin levels may be helpful in ruling out factitious insulin use, however note the limited reactivity of various insulin analogs. Accordingly, one use of the insulin test is to screen patients with low glucose for insulinoma if other factors such as glucose-lowering drugs or reactive hypoglycemia are excluded.

Insulin levels can be also useful in predicting susceptibility to the development of type II diabetes.⁴ However, the American Diabetes Association recommendations for the diagnosis of diabetes do not include the measurement of insulin levels.⁶

Insulin levels may also be measured to estimate the patient's capacity for insulin secretion (eg, glucagon test) or in the evaluation of insulin sensitivity (eg, oral glucose tolerance test). Although the adequacy of pancreatic insulin synthesis is frequently assessed via the determination of C-peptide, fasting insulin is also measured to assess insulin resistance.

In treated patients therapeutic administration of insulin can lead to the formation of insulin binding antibodies. Insulin binding antibodies invariably interfere with insulin measurement methods and results are unreliable. For patients with known autoantibodies to insulin should measure free and total insulin to assess biologically active fraction of insulin (free insulin) and total insulin. Measurements of free and total insulin, C-peptide, insulin antibodies and hemoglobin A1C are often tested in conjunction to clarify the contribution of endogenous and exogenous insulin to overall diabetic management. See Test **501561** (/tests/501561/insulin-free-and-total) for Free and Total Insulin.

Specimen Requirements

Specimen

Serum

Volume

0.8 mL

Minimum Volume

0.3 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube or gel-barrier tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube. Avoid hemolysis.

Stability Requirements

Temperature	Period
Room temperature	1 day
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

2.6–24.9 μ IU/mL

Storage Instructions

Refrigerate.

Patient Preparation

Patient should be fasting at time of collection.

Causes for Rejection

Citrate plasma specimen; hemolyzed specimen

References

Chevenne D, Letailleur A, Trivin F, Porquet D. Effect of hemolysis on the concentration of insulin in serum determined by RIA and IRMA. *Clin Chem*. 1998 Feb; 44(2):354-356. PubMed 9474041

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Footnotes

1. Insulin on Elecsys 1010/2010 and Modular Analytics E170, package insert 2006-11, V 7, Indianapolis, IN: Roche Diagnostics, 2006.
2. Owen WE and Roberts WL. Cross-reactivity of three recombinant insulin analogs with five commercial insulin immunoassays. *Clin Chem*. 2004; 50 257-259. PubMed 12502614 (<http://www.ncbi.nlm.nih.gov/pubmed/12502614>)
3. Merrigan, SD, Owen, WE, Laùlu1, SL, Wyness, SP, Roberts, WL, Straseski, JA. Cross-reactivity of modern insulin analogs insulin detemir and insulin glulisine with six automated insulin immunoassays. *ENDO*. 2012 poster PubMed 10505204 (<http://www.ncbi.nlm.nih.gov/pubmed/10505204>)
4. Okabayashi T, Shima Y, Sumiyoshi T, Kozuki A, Ito S, Ogawa Y, Kobayashi M, Hanazaki K. Diagnosis and management of insulinoma. *World J Gastroenterol*. 2013 February 14; 19(6): 829-837.

5. Sacks DB, Arnold M, Bakris GL, Bruns DE, Horvath AR, Kirkman MS, Lernmark A, Metzger BE, Nathan DM. Executive summary: guidelines and recommendations for laboratory analysis in the diagnosis and management of diabetes mellitus. *Clin Chem*. 2011 Jun; 57(6):793-798.
6. American Diabetes Association. Standards of Medical Care in Diabetes - 2014. Diabetes Care Volume 37, Supplement 1, January 2014.